Exp No: 8	**NMAP to Discover Live Hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform**

AIM:

To use the Nmap tool on the TryHackMe platform to discover live hosts in a network using ARP, ICMP, TCP and UDP ping scans.

ALGORITHM / PROCEDURE:

1. Log in to the TryHackMe website, open the *Nmap Live Host Discovery* room and click *Start AttackBox*. Click *Start Machine* to deploy the target.
2. Open the terminal in the AttackBox and note the IP address of the AttackBox using `ip a`.
3. ARP scan: run `nmap -PR -sn 10.10.5.0/24`. ARP works only for hosts on the same subnet.
4. ICMP scan: run `nmap -PE -sn 10.10.5.0/24` to send ICMP echo requests (ping).
5. TCP scan: run `nmap -PS22,80,443 -sn` (SYN ping) and `nmap -PA80 -sn` (ACK ping).
6. UDP scan: run `nmap -PU53,161 -sn` to send empty UDP packets to the given ports.
7. Note the hosts reported as "up" and compare the results of the four scan types.

Fig 1: Host discovery probes sent from the AttackBox

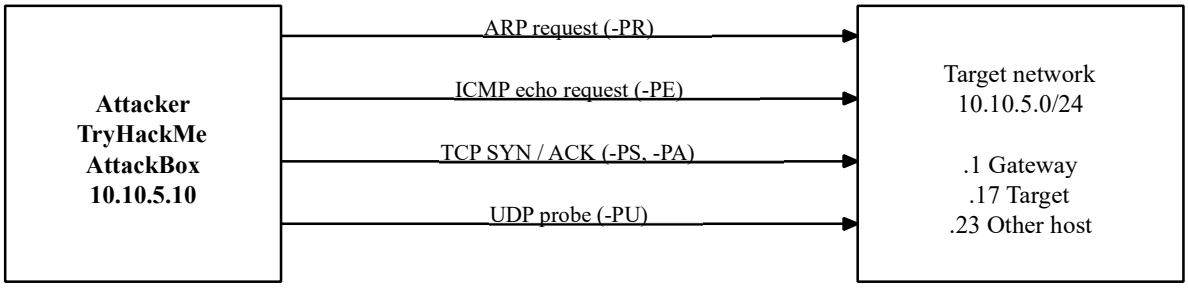
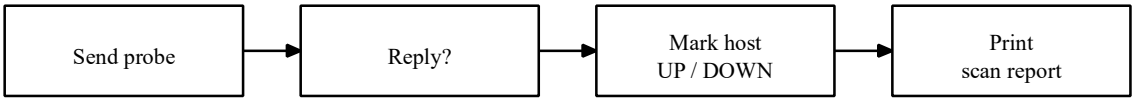


Fig 2: Working of a host discovery scan



Commands used:

Scan type	Option	Command
ARP ping	`-PR`	`nmap -PR -sn 10.10.5.0/24`
ICMP echo ping	`-PE`	`nmap -PE -sn 10.10.5.0/24`

TCP SYN ping	`-PS`	`nmap -PS22,80,443 -sn 10.10.5.0/24`
TCP ACK ping	`-PA`	`nmap -PA80 -sn 10.10.5.0/24`
UDP ping	`-PU`	`nmap -PU53,161 -sn 10.10.5.0/24`
List scan (no probes)	`-sL`	`nmap -sL 10.10.5.0/24`

OUTPUT:

AttackBox terminal - ARP scan

```
root@ip-10-10-5-10:~# nmap -PR -sn 10.10.5.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-07 10:42 BST
Nmap scan report for 10.10.5.1
Host is up (0.00031s latency).
MAC Address: 02:2B:6C:41:9A:01 (Unknown)
Nmap scan report for 10.10.5.17
Host is up (0.00045s latency).
MAC Address: 02:7E:D3:08:5C:17 (Unknown)
Nmap scan report for 10.10.5.23
Host is up (0.00052s latency).
MAC Address: 02:19:A4:F2:70:23 (Unknown)
Nmap scan report for 10.10.5.10
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.87 seconds
```

AttackBox terminal - ICMP echo scan

```
root@ip-10-10-5-10:~# nmap -PE -sn 10.10.5.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-07 10:44 BST
Nmap scan report for 10.10.5.1
Host is up (0.00029s latency).
MAC Address: 02:2B:6C:41:9A:01 (Unknown)
Nmap scan report for 10.10.5.17
Host is up (0.00041s latency).
MAC Address: 02:7E:D3:08:5C:17 (Unknown)
Nmap scan report for 10.10.5.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.11 seconds
```

AttackBox terminal - TCP SYN ping

```
root@ip-10-10-5-10:~# nmap -PS22,80,443 -sn 10.10.5.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-07 10:46 BST
Nmap scan report for 10.10.5.17
Host is up (0.00038s latency).
MAC Address: 02:7E:D3:08:5C:17 (Unknown)
Nmap scan report for 10.10.5.23
Host is up (0.00049s latency).
MAC Address: 02:19:A4:F2:70:23 (Unknown)
Nmap scan report for 10.10.5.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.04 seconds
```

AttackBox terminal - TCP ACK ping

```
root@ip-10-10-5-10:~# nmap -PA80 -sn 10.10.5.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-07 10:47 BST
Nmap scan report for 10.10.5.17
Host is up (0.00036s latency).
MAC Address: 02:7E:D3:08:5C:17 (Unknown)
Nmap scan report for 10.10.5.10
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.02 seconds
```

AttackBox terminal - UDP ping

```
root@ip-10-10-5-10:~# nmap -PU53,161 -sn 10.10.5.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-07 10:49 BST
Nmap scan report for 10.10.5.17
Host is up (0.00044s latency).
MAC Address: 02:7E:D3:08:5C:17 (Unknown)
Nmap scan report for 10.10.5.10
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.15 seconds
```

RESULT:

Thus the live hosts in the network were discovered using Nmap with ARP, ICMP, TCP and UDP scans on the TryHackMe platform. The ARP scan found the most hosts (4) because it works at the data link layer and cannot be blocked by a firewall on the same subnet, while the ICMP, TCP and UDP scans depended on how the target responded to each probe.